



Society of Petroleum Engineers

SPE-227125-MS

Standard Petrography and Digital Rock Physics as Tools for Rock-Typing Studies. Example of the Microporous Dolomitized Intervals of the Buwaib Formation (Valanginian), Saudi Arabia

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227125-MS](https://doi.org/10.2118/227125-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

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Abstract

In carbonate rocks, establishing rock-types that honor both geology and petrophysics is always a challenge. This is mainly due to intrinsic pore system heterogeneity and sensitivity to diagenetic alterations. Furthermore, some routine and/or special core analyses are destructive and time consuming. Over the last decade, non-destructive micro computed tomography supported the development of digital rock physics techniques, enabling pore system observation and fluid flow modeling, with high resolution and limited to no damage to the rock samples.

The studied interval includes the Valanginian (Lower Cretaceous) Buwaib Formation of Saudi Arabia. In this study, standard petrography techniques such as thin-section observations and Scanning Electron Microscopy (SEM) were integrated with state-of-the-art micro-computed tomography observations and digital rock physics simulations. Those data were further compared with plug-scale measurements such as Mercury Injection Capillary Pressure (MICP) data to validate the models. This workflow was designed in order to establish a rock-based rock-typing scheme that would integrate both geology and petrophysical data and enable refined hydrocarbon saturation estimates.

Studied deposits include intertidal mudstones and wackestones, usually pervasively dolomitized microporous. Diagenetic studies reveal that dolomite formed at an early stage with limited influence of burial diagenesis. Under SEM, dolomite fabrics are very finely crystalline with euhedral rhombs. Micropores are abundant and well connected, but the presence of detrital clays locally increases the tortuosity of the pore system suppressing connectivity. Based on those observations, the integration of the CCA and MICP data lead to the characterization of several rock-types with their own petrographic and petrophysical characteristics. A comprehensive digital rock physics workflow was utilized to produce 3D rock models through multi-scale 3D X-ray microscopy, and application of a machine-learning algorithms, to characterize the internal fabric of the rock samples based on the abundance of microporosity. Numerical simulation of the electrical current flow through the samples showed the variation of Archie's saturation exponent "n" with water saturation. The meso- and macropores were found to produce the moveable hydrocarbons due to their lower capillary pressure, whilst the micropores hold immobile formation water. The digital models

were then used to simulate the drainage of the relative permeability to identify water saturation changes affecting the timing of first water flow in subsurface.

Multiphase flow and electrical properties derived from digital experiments indicate that pore structure and sizes are major controls on them in this carbonate formation. Variable saturation exponent values indicate the Archie model may not be appropriate for resistivity log-based water saturation calculation. Saturation calculation could be significantly refined, if wireline petrophysical models are calibrated using digital rock analysis data.

Introduction

Microporosity is characterized by pores smaller than 10 μm in diameter and usually with pore throat diameter less than 0.5 μm is variably abundant in carbonate strata both in subsurface and outcrops. These small pores are developed either within the micritic matrix or in the allochems, especially within limestones, but can also be present in dolomite fabrics. Owing to their small sizes, micropores, even where abundant, suppress permeability and are also associated with high bound water, reducing resistivity readings and impairing proper reservoir characterization. The characterization of the microporous network in limestones has been the focus of a literature over the last decades, concentrating either on the characterization of the pore system through petrophysical analyses, or on the sedimentological and diagenetic origin of the microporous systems. However, classical petrographic tools such as thin sections do not have the resolution to directly observe the micropore systems and therefore more advanced tools such as SEM or confocal microscopes are needed. Plug-scale conventional core analyses (CCA) and Mercury Inject Capillary pressure (MICP) data are also usually implemented to characterize the pore volumes, along with their distribution at the plug scale. These techniques have the major disadvantage of being destructive and to be of limited use for flow simulation and 3D modeling of the pore volume, although provide relatively fast and cost-efficient observations. In this context, the development of Digital Rock Physics (DRP), driven by increasing computation capabilities and CT-scan resolution provides a very good tool to visualize the pore volume in 3D, but also and most importantly to replicate CCA measurements, simulate fluid flow with appropriate input parameters, and ultimately to mimic special core analyses measurements (SCAL), and all of these without damaging the samples. In the case of microporous carbonates, DRP analyses appear to be a major step forward in the characterization of the pore volumes at the plug-scale and should be very helpful for rock typing studies and flow simulations. This study combines both petrographic and DRP approaches to provide a detailed characterization of the pore system within a microporous dolomite unit from the Buwaib Formation (Valanginian) of Saudi Arabia, with the aim to provide an integrated rock-typing analysis that will support refinement of hydrocarbon saturation.

Geological background

During the Early Cretaceous, the Arabian Platform was characterized by a large, 100s-km wide, epeiric platform located on the passive margin of the Tethys Ocean (Ziegler, 2001 – Figure 1). Deposits are dominated by shallow marine carbonates prograding towards the Tethys margin located in modern Oman (Ziegler, 2001). The studied section, located in the Rub al Khali Basin (Saudi Arabia) encompasses the lowermost half of the Buwaib Formation (Thamama Group) and is the direct regional time equivalent of the Zakum Member in the UAE (part of the Lekhwair Formation as defined by Sharland et al., 2001; Granier, 2008), and of the Habshan Formation in Oman. During this time, the Arabian Platform constituted a large, shallow marine, carbonate system showing a broad southwest /northeast progradation (Figure 1C), marked locally by low-angle clinoforms, especially in Oman and on the eastern side of the UAE (e.g., Dujoncquoy et al., 2018). In the UAE and in Saudi Arabia, the deposits of the Buwaib Formation are broadly characterized by alternating moderate-energy inner ramp facies, locally rich in *Lithocodium*/*Bacinella* lumps, and argillaceous low-energy inner ramp sediments (e.g. Bourdarot et al., 2014; Khemissa et

al., 2016; Jeong et al., 2016, 2017; Tendil et al., 2021; Deville de Periere et al., 2023). The deep marine facies rich in calpionellids described in Oman by Wohlwend et al. (2016) and Celestino et al. (2017) represent the most distal deposit of the Valanginian described on the Arabian Platform.

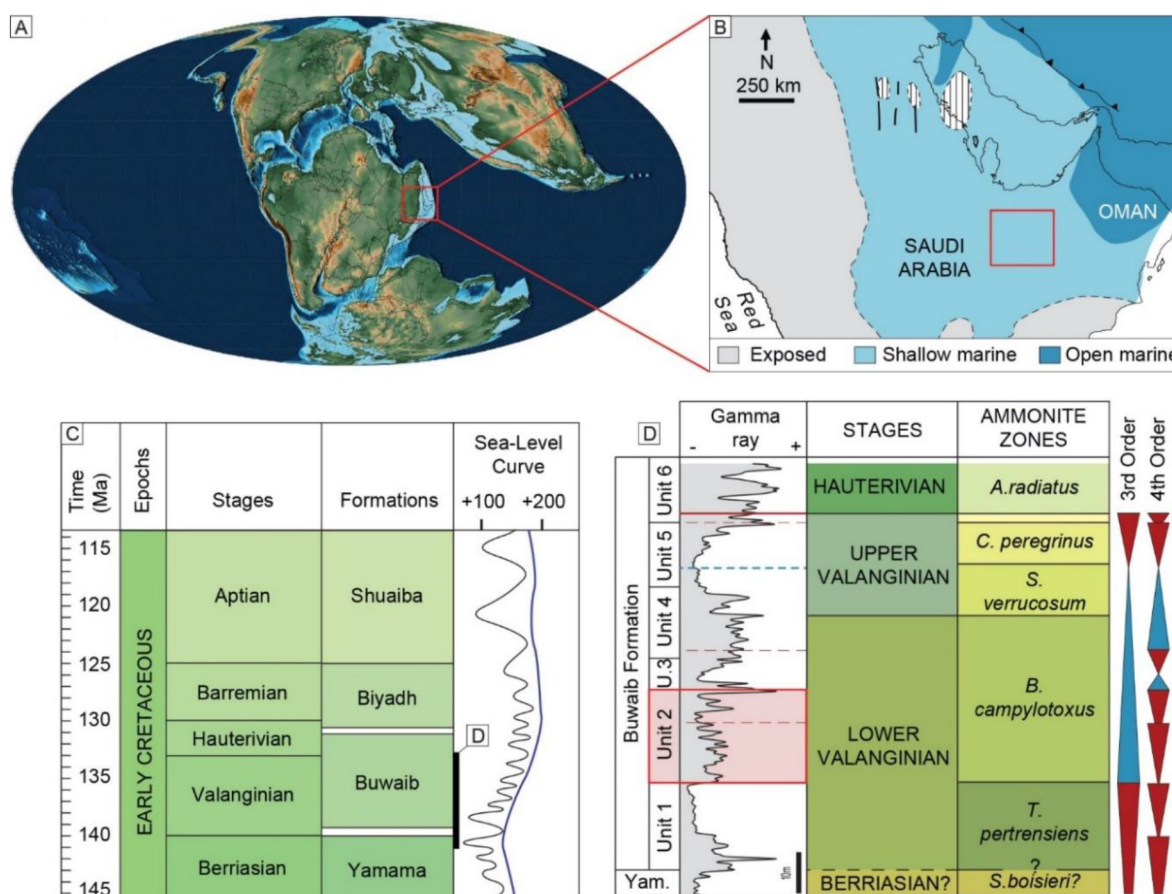


Figure 1—A. Global paleogeographical map of the Valanginian (Scotese, 2014); (B) Simplified paleoenvironmental map of the Arabian platform during the Valanginian (modified from Ziegler, 2001); (C) Early Cretaceous stratigraphy of Saudi Arabia including eustatic variations of the Arabian Platform (modified from Haq & Al-Qahtani, 2005) (D) Detailed stratigraphy of the Buwaib Formation in the Rub al Khali Basin (modified from Deville de Periere et al., 2023).

Sedimentology, diagenesis and rock-typing

Sedimentological make-up

The studied section corresponds to the Unit 2 of the Buwaib Formation and accounts for the lower half of the lower Valanginian *B. campylotoxus* biozone (Deville de Periere et al., 2023 – Figure 1). This unit is approximately 20m-thick and is typically dominated by intertidal dolomitized mudstones. Sedimentary structures include fenestrae, rootlet traces, rare teepees and possible ripple marks. Bioturbation is locally pervasive, mainly including undefined horizontal burrows. Locally, microbial bindstones are affected by desiccation and disrupted by brecciated surfaces and common intraclasts and minor anhydrite nodules. In those sections, only rare biocomponents have been observed, including microbial debris and laminations and rare ostracods. The dolomitized mudstones have a chalky texture and are locally associated with greenish clays dominated by illite and possible palygorskite, with subordinate detrital quartz/k-feldspars. The dolomitized intertidal deposits change basinward to low-energy inner ramp, skeletal wackestones and packstones that are only sporadically dolomitized in the vicinity of high frequency sequence boundaries (Deville de Periere et al., 2023). In these more open marine facies, skeletal allochems include minor green

algae (dominated by *Salpingoporella annulata*), gastropods, bivalve debris, ostracods and rare benthic foraminifera (*Lenticulina*, *Everticyclammina kelleri*, miliolids).

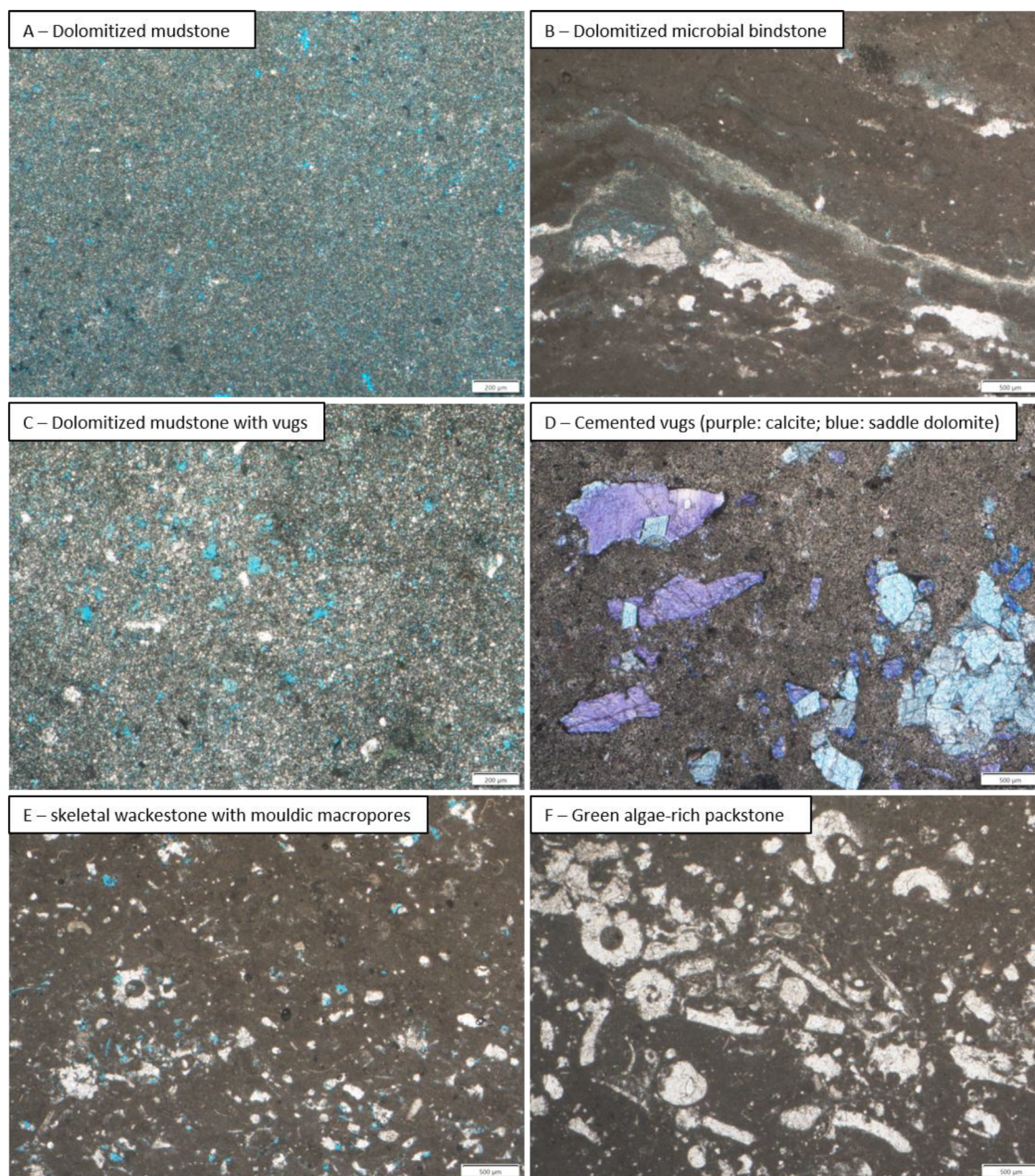


Figure 2—Representative thin-section photomicrographs illustrating the various microfacies and diagenetic overprint affecting the studied interval.

Diagenetic overprint

In the studied section, dolomitization is pervasive, especially in the intertidal deposits. The replacive dolomite is fabric preserving, fine to very fine-crystalline and SEM observations show that rhombs are typically euhedral, representing a ‘micro planar-e’ fabric. Stable isotopes analyses on the dolomite matrix suggest limited recrystallization during burial ($\delta^{18}\text{O}$ comprised between -1‰ and +3‰ V-PDV), while the negative $\delta^{13}\text{C}$ values reaching -4.5‰ might mark the incorporation of organic carbon derived from paleosoils. In the dolomitized sections, macropores and fractures are occluded by coarse-crystalline ferroan

saddle dolomite and blocky weakly ferroan calcite cements, both of these phases being related to burial diagenesis.

Controls porosity development and core-based rock-typing

Core observations, integrated with thin-sections, SEM, CCA data (ie. porosity / permeability) and MICP data, provide insights on the rock-typing of the studied interval, which are detailed below and illustrated in Figure 3.

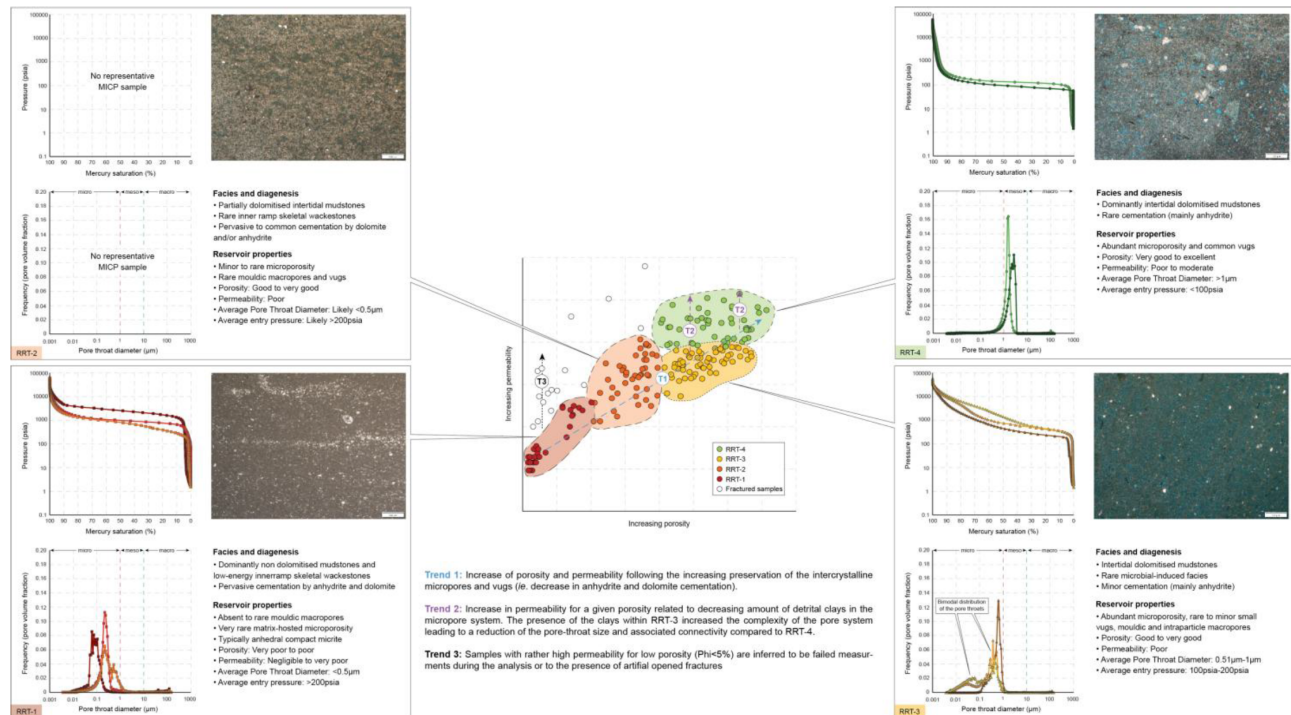


Figure 3—Rock-typing scheme integrating core, thin section, CC and MICP data.

In the studied interval, there is clear dichotomy between the dolomitized units and the limestone beds in terms of pore systems. Indeed, the visible pore volume in the dolomitized sections is dominated by variably abundant micropores, along with rare small vugs. In these facies, the coeval increase of porosity and permeability is directly related to the preservation of the microporous system, along with the increase in non-cemented small vugs. According to SEM observations and XRD data, illitic clays are present in variable amounts, which in turn is also a direct factor controlling permeability: for a given porosity, permeability appears to be up to two orders of magnitude lower in samples that show minor to common illite within the micropore systems. Given the size of the clays, porosity is barely modified, but the presence of the clays in the pores creates a much higher tortuosity of the pore systems and decreases permeability. In the non-dolomitized sections, matrix-hosted microporosity is less developed and solution-related macropores such as molds and vugs are more abundant, along with subordinate primary intraparticle macropores. Under the SEM, the micritic matrix in the limestone varies between anhedral compact and more rarely polyhedral to sub-rounded (*sensu* Deville de Periere et al., 2011).

The core-based rock-typing presented in this study is therefore as follow:

- RRT-1 corresponds to low-energy inner ramp, non-dolomitized, deposits that are pervasively cemented. In this rock-type, macropores are pervasively cemented by calcite and dolomite, and the micritic matrix is characterized by an anhedral compact fabric with only a limited amount of

isolated micropores. Hence, the average pore-throat diameter is very small, usually $<0.5\mu\text{m}$ with relatively narrow distributions of the pores.

- RRT-2 comprises partially dolomitized intertidal mudstones along with rare low-energy inner ramp non-dolomitized skeletal wackestones. Macropores are also pervasively cemented by calcite and dolomite, but matrix-hosted micropores are rare to common, usually patchily distributed within the samples (at least at the plug scale). Although there are no MICP data associated with this rock-type, an average pore throat diameter $<0.5\mu\text{m}$ is inferred given the limited measured permeability.
- RRT-3 encompasses fully dolomitized and micropores intertidal mudstones with small euhedral dolomite rhombs. Macropores are usually occluded by saddle dolomite with subordinate calcite. The micropore system is well-developed but the presence of illite in the micropores reduces the overall connectivity of the pore system. Hence, pore throat diameters range from $0.5\mu\text{m}$ to $1\mu\text{m}$ and show a wide unimodal to locally bimodal distribution suggesting a plug-scale heterogeneity in the pore size and connectivity.
- RRT-4 is the best target rock type observed in the studied section and only differs from RRT-3 by the presence of slightly more abundant vugs, and by a drastic decrease in clay abundance. Samples observed in this rock-type have average pore throat diameters $>1\mu\text{m}$ with a narrow unimodal distribution typical of microporous carbonates. The micropore system in this rock-type appears more homogeneous than in RRT3.

Digital rock physics

The microporous rock samples underwent comprehensive evaluation via Digital Rock Physics (DRP) analysis, which was subsequently validated through comparative assessment with multiple complementary methodologies, including MICP, petrographic analysis, SEM, and CCA. The resultant data exhibited satisfactory concordance, with reservoir rock typology, pore throat diameter, total porosity, and absolute permeability values derived from DRP, MICP, petrography, SEM, and RCA demonstrating strong mutual confirmation. The primary outcome of this dataset was the development of a calibrated DRP model, which was subsequently employed to simulate SCAL properties, including resistivity index and drainage-induced relative permeability for gas-water systems, via numerical modeling techniques.

A rigorous DRP workflow was employed to generate high-fidelity 3D digital rock models of the rock samples via multi-resolution 3D X-ray CT-imaging and voxel-based numerical simulations of the petrophysical properties, thereby facilitating a detailed characterization of the internal rock fabric and its associated microporosity distribution. The 50% of original MICP datasets (20 plugs) served as a calibration framework for the 3D digital rock models, enabling the establishment of a robust and accurate representation of the rock samples' internal structure and petrophysical properties.

Workflow

The workflow comprises a multi-step protocol, encompassing the following components:

1. Petrographic analysis of thin-sections to elucidate lithofacies and pore type distributions, thereby providing a foundational understanding of the rock's mineralogical and textural characteristics that serve as a foundation of the DRP segmentation algorithms.
2. Pore structure analysis and clustering utilizing MICP data, which enables the identification of distinct pore networks and their associated morphological attributes as twin-plugs selection for the digital imaging.
3. Multi-scale 3D X-ray microscopy of the selected rock samples at voxel sizes varying from $10\text{--}20\mu\text{m}$ for 1-1.5 inch full plugs down to $0.5\mu\text{m}$ for high-resolution sub-plugs and sub-volumes ranging in size from 3mm to 0.5mm and representing individual rock types.
4. Segmentation of the resultant 3D images into pores and grains of different mineralogy.

5. Simulation of static (porosity, connected and isolated) and dynamic (permeability, absolute and relative, resistivity index, etc.) rock properties.
6. Establishment of linkages between rock types and petrophysical properties, simulated from micro-computed tomography (microCT) scans, to elucidate the controls exerted by pore systems on petrophysical behavior, including permeability, porosity, and fluid flow properties.
7. Generation of a comprehensive pore structure and petrophysical properties database for Low-Resistivity Pay (LRP) rocks, which serves as a critical input for downhole well log petrophysical models, enabling the accurate prediction of reservoir properties and optimization of hydrocarbon discovery.

The end-goal of this process is to evaluate cementation exponents ‘m’, saturation exponents ‘n’, water-gas relative permeabilities and capillary pressures of each RRT.

Methodology and Results

A comprehensive rock imaging protocol was employed to acquire high-resolution (10-20 $\mu\text{m}/\text{voxel}$) scans of sixty-two full core plugs, facilitating the identification and characterization of rock types and their spatial distribution within the core plugs, as well as the selection of optimal locations for micro-plug extraction. Subsequently, twenty micro-plugs were extracted and subjected to high-resolution (0.5 $\mu\text{m}/\text{voxel}$) imaging, followed by image processing and analysis, yielding 3D digital rock models (see Figure 4 for an exemplary representation).

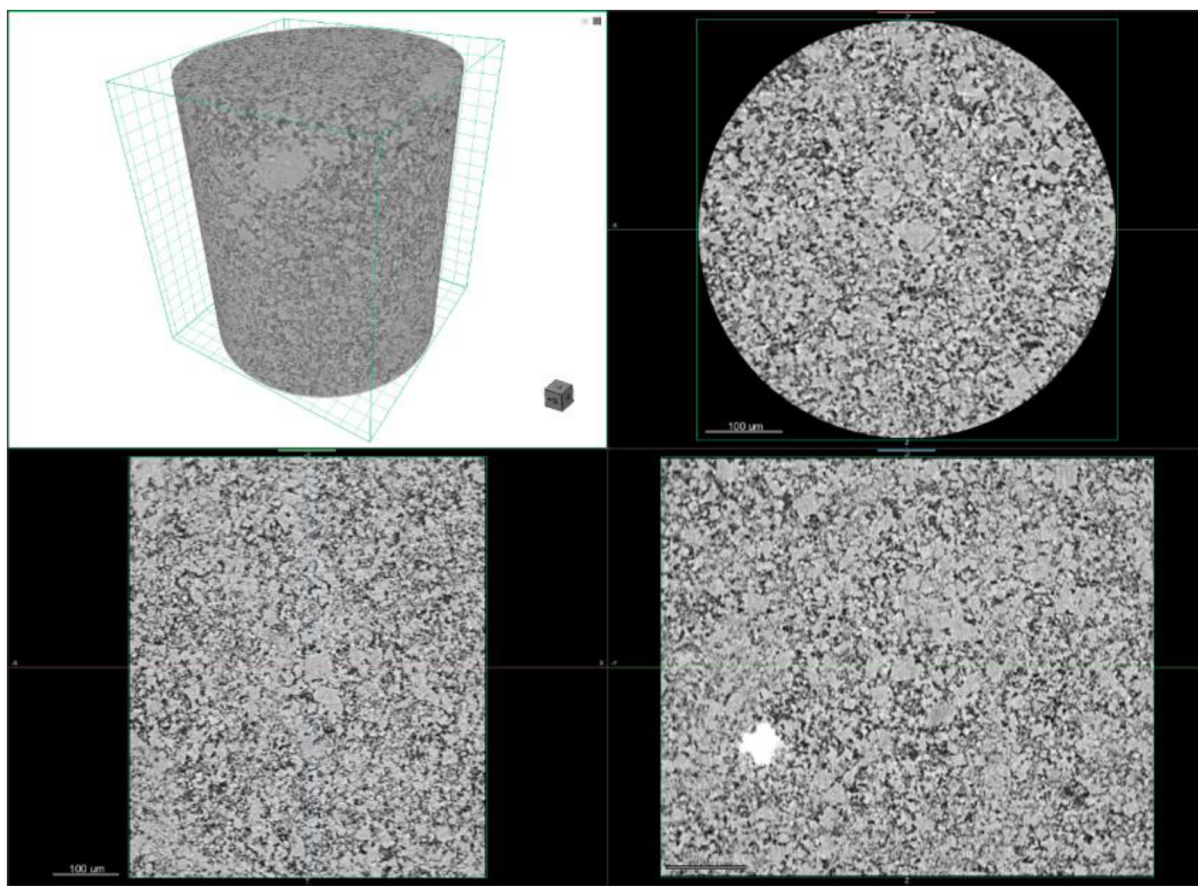


Figure 4—Typical microCT image of the microporous carbonate sample: three orthogonal projections and 3D render. Voxel size 0.46 μm .

Pore space segmentation was achieved through hybrid thresholding and watershed transform algorithm to the gradient tomography data. The resultant pore separation was obtained via a geometrical watershed transform algorithm, enabling the analysis of pore body volumes and geometries. Pore throats were extracted, and a pore size distribution was computed (Figure 5A), which was subsequently validated against pore throat size distributions derived from MICP experiments.

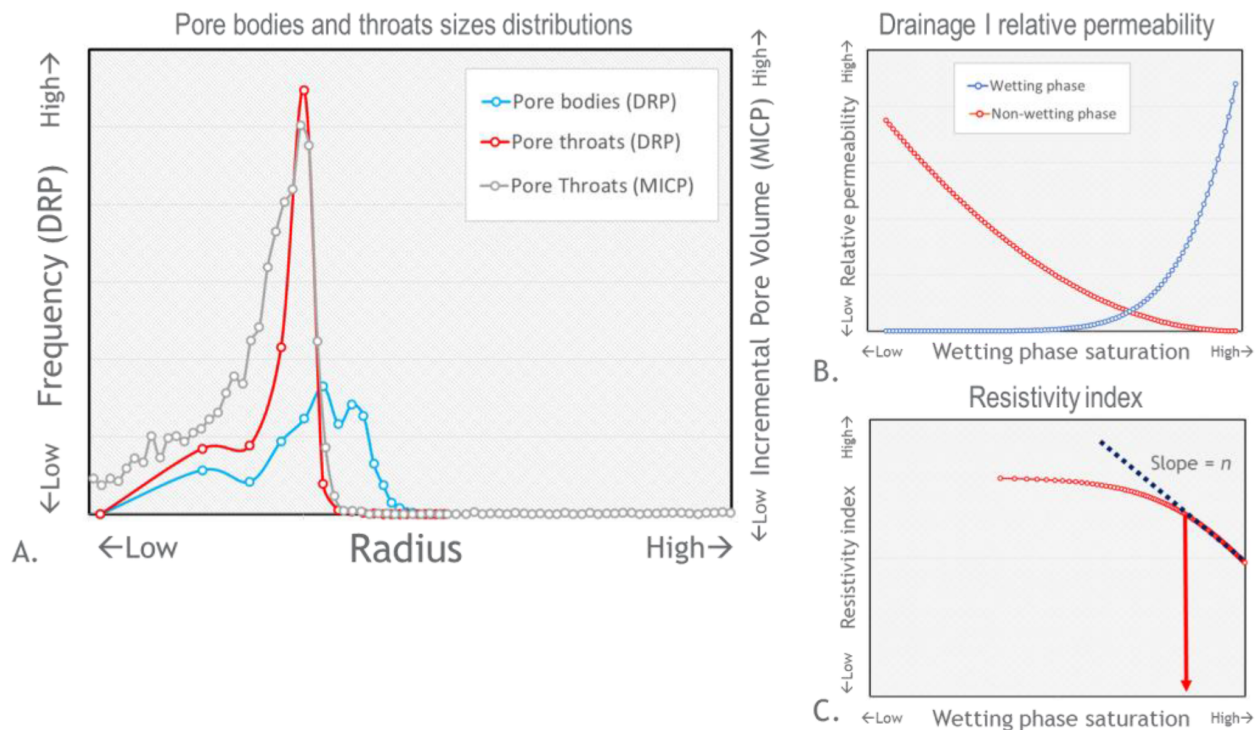


Figure 5—Digital simulation results for one of the carbonate microporous sample: A) pore throat and body sizes radius distributions from DRP compared to MICP dataset; B) drainage I relative permeability curves for wetting phase (water) and non-wetting phase (gas), steady state experiment simulation; C) resistivity index curve simulated for the gas-water system.

The petrophysical properties of both the intergranular and microporous pore spaces were quantified through the numerical solution of the Navier-Stokes equation for absolute permeability (Toelke, et al., 2010). Furthermore, a random walk algorithm was employed to solve the Laplace equation for Formation Resistivity Factor (FRF), facilitating the derivation of the cementation exponent "m". This integrated approach enabled the comprehensive characterization of the rock's petrophysical properties, providing valuable insights into the complex relationships between pore structure, fluid flow, and electrical conductivity.

The capillary pressure curve was computed utilizing a pore morphology method-based algorithm, specifically the "maximum inscribed spheres" approach (Silin & Patzek, 2006), which leverages the distribution of two immiscible fluids to derive the capillary pressure-saturation relationship. The outcome of this pore morphology method is a sequence of quasi-stationary two-phase distributions, which serve as the basis for calculating the saturation-dependent capillary pressure curve.

Subsequently, the relative permeability was simulated by solving the Navier-Stokes equation for the pore space, wherein the individual fluid phases are connected at each water saturation level (Arnold et al., 2003). This approach restricts the two-phase flow to the connected phase volume, thereby enabling the accurate prediction of relative permeability. An illustrative example of relative permeability simulation is presented in Figure 5B.

The resistivity index was computed for the same fluid phase distribution steps as the relative permeability, with the caveat that such simulations necessitate the definition of pore wall wettability. To this end, the

wettability of the pore walls was measured experimentally using the U.S. Bureau of Mines (USBM) method for analogous rock types in nearby legacy wells. The carbonate rocks under investigation were deemed to be primarily water-wet, as evidenced by the experimental results.

The relative permeability curves, as well as the resistivity index curves and corresponding saturation exponents, were established for gas drainage at initial water saturation (S_{wi}), as depicted in Figures 5B and 5C. These curves provide a comprehensive characterization of the rock's petrophysical properties, including its capillary pressure, relative permeability, and resistivity index, which are essential for predicting fluid flow and hydrocarbon recovery in carbonate reservoirs.

Conclusions

A comprehensive digital rock modeling and pore system characterization workflow was employed to create and refine formation-specific rock types, thereby providing a robust foundation for subsequent simulations and analyses. The methodology presented in this study is specifically designed to support drilling operations by providing refined electrical and multi-phase flow models, which enable the identification of sweet spots in low resistivity pay zones. This approach is distinct from field development planning, as operational decisions must be made within a short timeframe that precludes the execution of complex petrophysical and imaging experiments. The simulation results are presented and discussed below.

The pore size distribution and geometry of the studied formation introduce significant variations in the cementation factor and saturation exponent, which are critical parameters in petrophysical modeling. The presence of microporosity is found to dominate resistivity measurements, necessitating the development of a dual porosity rock digital model to accurately describe this complex pore system. The results demonstrate that the abundance of micropores can have a profound impact on the computed values of saturation from electrical wireline logs and relative permeability, highlighting the importance of incorporating microporosity into petrophysical models.

The simulated cementation factor (m) and saturation exponent (n) values for the studied samples exhibit a range of 1.7 to 2.0 and 1.7 to 1.9, respectively. Notably, samples with distinct pore models display unique m and n values, underscoring the importance of accurate pore system characterization. Furthermore, variable saturation exponent values are observed, which suggests that the Archie model may not be universally applicable for calculating water saturation from electrical wireline logs.

These observations indicate that in the studied dolomitized formation, porosity and permeability evolution are primarily controlled by diagenetic processes (e.g., dolomitization), with changes in the abundance of intercrystalline microporosity and detrital kaolinite clays playing a crucial role. The combination of microporosity and clays provides a plausible explanation for the low resistivity of this formation, which is attributed to the presence of abundant bound water.

Acknowledgements

Authors appreciate Dr Hongwen Zhao, Dr Ming Zhang and Dr Yufeng Cui, geological scientists of Aramco Beijing Research Center for technical discussions.

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